

Problem Statement and Goals

FPVR

Team 9, Software Mandems

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Table 1: Revision History

Date	Developer(s)	Change
Sept 27, 2026	John Cho	Initial Problem Statement
Sep 27 2026	Vikram Chandar	Goals, Stretch Goals and Environment

1 Problem Statement

1.1 Problem

Learning to fly a FPV (First person view) drone can be difficult and unsafe especially for beginner pilots. If left unaccounted for, they can risk damaging their drones, nearby property, or other equipment while learning basic flight controls and maneuvers.

Practicing and flying in general through traditional means can also be limited by the need for large open spaces, weather conditions, and access to costly hardware. These barriers can make it difficult for anybody to fly a drone with little to no consequences.

1.2 Inputs and Outputs

Inputs:

- Joystick controller or similar interface (over RF or Serial)

Outputs:

- Simulated feedback from the VR
- AI-integrated performance analysis at the end of a course/session

1.3 Stakeholders

1.3.1 Primary Stakeholders

- **Drone Pilots:** FPVR will be designed with beginner to intermediate level pilots in mind, licensed or not, who have yet to familiarize themselves with flying drones in the field.
- **Drone Training Organizations:** In Canada, any drone above 250g in weight requires a pilot with a formal certificate to operate in a public space [1]. Realistic drone flight simulation may complement or outright replace the mandatory in-person flight exams facilitated by drone schools.

1.3.2 Secondary Stakeholders

- **Drone Clubs and Teams:** Local technical clubs designing and building teleoperated FPV drones can benefit from FPVR and provide quick and useful feedback for improving the product.
- **FPV Communities:** Anyone from a passive to hardcore FPV hobbyist within a drone community might find this technology useful both for casual and practical use.

1.4 Environment

1.4.1 Hardware Environment

The simulator will run on a compatible VR headset capable of supporting real-time 3D applications. The headset will provide the display and motion tracking required for the user to interact with the simulated environment. VR controllers or compatible FPV flight controls will be used to control the simulated drone.

1.4.2 Software Environment

The simulator will run on the VR headset using a VR-compatible operating system and runtime environment. The software will require the necessary VR support to provide real-time graphics, motion tracking, and user input. The simulator will operate as a standalone VR application on supported hardware.

2 Goals

1. **Provide a Realistic FPV Drone Flight Experience:** Create an immersive virtual reality FPV drone simulation that maintains a smooth and consistent frame rate during normal gameplay.

2. **Support Beginner and Advanced Flight Training:** Enable new users to develop fundamental FPV flight skills while allowing experienced pilots to refine their performance through improvements in course completion times, flight consistency, and crash frequency.
3. **Provide Realistic and Responsive Flight Controls:** Deliver intuitive controls with minimal input delay and realistic, consistent drone responses to throttle, pitch, roll, and yaw inputs.
4. **Provide Varied Flight Environments and Courses:** Offer multiple environments and courses featuring different layouts, obstacles, and flight challenges to provide varied training scenarios.
5. **Track Flight Performance:** Record basic flight performance data, including completion times, crashes, and course completion status, allowing users to review and monitor their progress.

3 Stretch Goals

1. **Provide Multiple Drone Configurations:** Offer multiple drone configurations with distinct flight characteristics, handling, and responsiveness, allowing users to experience different drone setups and adapt their flying techniques accordingly.
2. **Support Multiplayer Flight:** Enable multiple users to operate drones within the same virtual environment, allowing them to practice and interact together in shared training scenarios.
3. **Provide AI-Based Flight Analysis and Feedback:** Analyze controller inputs and flight performance to identify patterns such as inefficient inputs, inconsistent control, excessive corrections, and other areas for improvement, and provide personalized feedback to help users refine their flying skills.

4 Custom Documents (CDs)

The extras chosen are the Design Thinking Report and the Usability Report. The Design Thinking Report will explain the intent behind our high level decisions and direction behind our design. The usability report will show the results of usability testing while also showing the response received from users. Our goals and methods regarding the testing will be documented as well.

Appendix — Reflection

[Not required for CAS 741 —SS]

The purpose of reflection questions is to give you a chance to assess your own learning and that of your group as a whole, and to find ways to improve in the future. Reflection is an important part of the learning process. Reflection is also an essential component of a successful software development process.

Reflections are most interesting and useful when they're honest, even if the stories they tell are imperfect. You will be marked based on your depth of thought and analysis, and not based on the content of the reflections themselves. Thus, for full marks we encourage you to answer openly and honestly and to avoid simply writing "what you think the evaluator wants to hear."

Please answer the following questions. Some questions can be answered on the team level, but where appropriate, each team member should write their own response:

1. What went well while writing this deliverable?
2. What pain points did you experience during this deliverable, and how did you resolve them?
3. How did you and your team adjust the scope of your goals to ensure they are suitable for a Capstone project (not overly ambitious but also of appropriate complexity for a senior design project)?

References

- [1] T. Canada, "Flying your drone safely and legally," Transport Canada, Nov. 04, 2025. <https://tc.canada.ca/en/aviation/drone-safety/learn-rules-you-fly-your-drone/flying-your-drone-safely-legally>